

Software Requirements Specification

Pet Adoption & Shelter Management System

Problem Statement #57 | Media, Events & Community
PES University, Dept. of CSE — Software Engineering
Nishit D B (PES1UG24CS303)
Version 1.1

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Pet Adoption & Shelter Management System, a web portal for an animal shelter. It is intended for the project developers, testers and course evaluators, and is the baseline for design, implementation and testing.

1.2 Scope

The system tracks animal rescue medical intake histories, lets prospective adopters search the pet catalog and submit adopter questionnaires, lets shelter staff screen applicants and decide on applications through to a completed adoption, and manages foster home placement timelines.

Out of scope: payment of adoption fees (collected at the shelter outside the system), donations and accounting, veterinary billing, and shelter supply inventory.

1.3 Definitions

Term	Meaning
Adoption Applicant	A member of the public who has registered and logged in to apply for an animal.
Shelter Staff	Authenticated shelter employee who manages animals, applications and foster placements.
Intake record	Vaccinations and medical examination logged when an animal enters the shelter.
Quarantine flag	Marker on an intake record for a condition needing isolation; blocks 'Ready for Adoption' until cleared.
Questionnaire	Adopter form: housing type, household members, other pets and adoption experience (all required).
Open application	An application in 'Pending Review' or 'Approved'.
Foster placement	Temporary placement with a foster caregiver, with start date and expected end date.
FR / NFR / RTM	Functional requirement / Non-functional requirement / Requirements traceability matrix.

1.4 References

- 57_SE_Lab1_SE_Problem_Statements.pdf — problem statement #57
- 1-RE/Requirements_Table.docx and 1-RE/RTM.xlsx
- 7-UML-Diagrams/Use_Case_Diagram.pdf — use-case diagram
- 7-UML-Diagrams/Activity_Diagram.pdf — activity diagram for UC-02
- 2-Architectural-Diagram/Architecture_Diagram.pdf — architecture diagram
- 4-SRS-and-WBS/UseCase_Flow_UC02.pdf and PES1UG24CS303_Nishit_DB_Exception_Flow.pdf — UC-02 flows
- 4-SRS-and-WBS/WBS.xlsx — work breakdown structure
- 6-Software-Testing/Bug_Project-NishitDB_PES1UG24CS303-BPS#57.PDF — Jira bug board
- IEEE Std 830-1998 — structure guidance

2. Overall Description

2.1 Product Perspective

A standalone web portal for one shelter, with a public catalog, a logged-in area for applicants, and a staff-only area for shelter operations. Email delivery is the only external service (status notifications).

2.2 User Classes

User	Description	Use cases
Adoption Applicant	Public user; may browse without login, must log in to apply. Sees only their own applications.	UC-01, UC-02, UC-10
Shelter Staff	Authenticated employee with access to animal, application and	UC-03, UC-05, UC-06,

	foster records.	UC-07, UC-09, UC-10
--	-----------------	---------------------

2.3 Operating Environment

Web application used from current desktop and mobile browsers. The browser must support local storage, used to keep an unsent questionnaire draft during a connectivity failure. Server and database platform are chosen in the design phase.

2.4 Constraints

- The lab scope is exactly five functional and two non-functional requirements (FR-001 to FR-005, NFR-001, NFR-002).
- Only animals with a complete intake record and no open quarantine flag may be adopted (FR-001).
- At most one application per animal may be Approved at a time (FR-004).
- Medical, background-check and questionnaire data are access-controlled (NFR-002).

2.5 Assumptions and Dependencies

- Background and reference checks are carried out by staff outside the system; only the results are recorded.
- An email service is available for notifications.
- The 200-concurrent-user peak load in NFR-001 is an assumed figure, to be confirmed with the shelter.

3. Use Cases

The use-case diagram is in 7-UML-Diagrams/Use_Case_Diagram.pdf; the UC-02 activity diagram is in 7-UML-Diagrams/Activity_Diagram.pdf.

ID	Use case	Actor(s)	Description	Requirements
UC-01	Search & Filter Pets	Adoption Applicant	Browse and filter the public catalog of animals that are 'Ready for Adoption'.	FR-002, NFR-001
UC-02	Submit Adoption Application	Adoption Applicant	Complete the adopter questionnaire for one animal and send it to the staff review queue.	FR-003, NFR-002
UC-03	Screen Adopter Questionnaire	Shelter Staff	Review a queued application's answers; always includes background verification (UC-04).	FR-004
UC-04	Verify Adopter Background	— (included by UC-03)	Record background and reference check results for the applicant.	FR-004, NFR-002
UC-05	Log Animal Medical Intake	Shelter Staff	Record vaccinations and medical examination; a complete record with no open quarantine flag makes the animal 'Ready for Adoption'.	FR-001, NFR-002
UC-06	Manage Foster Placement	Shelter Staff	Create, monitor (overdue list) and end foster placements; ending restores the animal's previous status.	FR-005
UC-07	Complete Adoption	Shelter Staff	At collection, mark an Approved application 'Completed'; the animal becomes 'Adopted'.	FR-004
UC-08	Send Status Notification	— (included by UC-07, UC-09)	Notify the applicant by email and in-app alert of an application status change.	FR-004
UC-09	Decide Application	Shelter Staff	Approve or reject a screened application, with a reason code on rejection.	FR-004
UC-10	Log In	Adoption Applicant, Shelter Staff	Register (applicants) and authenticate before using any non-catalog function.	NFR-002
UC-11	Place Animal in Quarantine	— (extends UC-05)	Open a quarantine flag when the intake examination finds a condition needing isolation; the animal cannot become 'Ready for Adoption' until staff clear it.	FR-001

3.1 Relationships

- UC-03 «include» UC-04 — Screening always includes background verification.
- UC-09 «include» UC-08 — Every decision notifies the applicant.
- UC-07 «include» UC-08 — Completing an adoption notifies the applicant.

- UC-11 «extend» UC-05 — Extension point 'examination finds a condition needing isolation'.

3.2 Core Use Case: UC-02 Submit Adoption Application

Related: Requires UC-10 Log In; follows UC-01 Search & Filter Pets; the created application is screened in UC-03 Screen Adopter Questionnaire. UC-02 has no include or extend relationships of its own.

Preconditions: The applicant is logged in (UC-10). The applicant has selected, from the catalog (UC-01), an animal whose status is 'Ready for Adoption' (FR-001).

Main success scenario:

1. Applicant selects "Apply to Adopt" on the animal's profile. The system displays the adopter questionnaire.
2. Applicant enters housing type, household members, other pets and adoption experience, and submits the questionnaire (FR-003).
3. System validates that all required fields are complete and well-formed.
4. System confirms that the animal is still 'Ready for Adoption' and that the applicant has no open application for it.
5. System creates the application with status 'Pending Review', links it to the animal and the applicant, and places it in the staff review queue.
6. System shows a confirmation with the application reference; the application appears under the applicant's "My Applications".
7. Use case ends successfully. Staff screening continues in UC-03.

Alternate flow 4a. Animal No Longer Available or Duplicate Application:

- 4a1. At step 4 the system finds that the animal has been removed from the catalog or is no longer 'Ready for Adoption' (e.g., another application was approved), or that the applicant already has an open application for it.
- 4a2. System does not create an application and explains why, showing the existing application if it is a duplicate.
- 4a3. System keeps the questionnaire answers so the applicant can reuse them for a different animal.
- 4a4. Use case ends; applicant may return to UC-01 Search & Filter Pets.

Exception flows:

E1. Invalid or Incomplete Input — Applicant leaves a required questionnaire field blank or enters malformed data (e.g., non-numeric household size). System detects the validation error at step 3 and displays the message: "Please complete all required fields before submitting your application.", highlighting the offending field(s) in red. System keeps the form populated with the applicant's existing entries and does not create an application record. Applicant corrects the flagged field(s) and resumes at step 2 of the Main Success Scenario.

E2. System / Connectivity Failure — Applicant submits the questionnaire, but the client cannot reach the backend or the backend cannot save the record to the database. System displays the message: "We couldn't submit your application due to a system error. Your answers have been saved on this device — we'll retry when the connection is back." The client (browser) stores the in-progress answers locally, since the server cannot save them, so no data is lost. No application record is created on the server. The failure is logged for Shelter Staff / IT review: by the backend if it was reachable, otherwise by the client, which reports it on reconnection. Once connectivity is restored, the client reloads the locally saved draft; the applicant reviews it and resumes at step 2 of the Main Success Scenario. If the server confirms the application had already been created (step 5) before the failure, the local draft is discarded and the flow continues at step 6.

Postconditions: Success: An application exists in state 'Pending Review', linked to the animal and the applicant, and is in the staff review queue (FR-003). The animal's status is unchanged. Failure: No application is created; the applicant's answers are kept on screen or as a local draft (see Alternate Flow 4a and Exception Flows E1, E2).

4. Functional Requirements

FR-001 — Medical intake and availability

Field	Detail
Priority	High
Description	The system shall allow staff to log animal vaccination and medical intake records, updating animal availability status to 'Ready for Adoption'.
Use case(s)	UC-05, UC-11
Acceptance criteria	Pass: Saving a complete intake record (vaccinations and medical examination) with no open quarantine flag sets the animal's status to 'Ready for Adoption'. Fail: The status becomes 'Ready for Adoption' while the intake record is incomplete or a quarantine flag recorded in it is still open.
Rationale	Given in problem statement — ensures only medically cleared animals become adoptable.

FR-002 — Catalog search and filter

Field	Detail
Priority	Medium
Description	The system shall allow an adoption applicant to search and filter the pet catalog by species, age, breed, and compatibility with children or other pets.
Use case(s)	UC-01
Acceptance criteria	Pass: Filtered results display only animals matching all selected criteria. Fail: Results include any animal not matching a selected filter.
Rationale	Main discovery function; Medium because applicants can still browse the unfiltered catalog.

FR-003 — Adopter questionnaire

Field	Detail
Priority	High
Description	The system shall allow a logged-in adoption applicant to submit an adopter questionnaire (housing type, household members, other pets, adoption experience — all required) for an animal whose status is 'Ready for Adoption'.
Use case(s)	UC-02
Acceptance criteria	Pass: A complete questionnaire creates an application in 'Pending Review', linked to the animal and applicant, that appears in the staff review queue within 5 seconds. Fail: An application is created with a required field missing, for an animal that is not 'Ready for Adoption' or has been removed from the catalog, or as a second open application by the same applicant for the same animal.
Rationale	Complete, structured adopter data is required before staff can screen an applicant.

FR-004 — Application decision workflow

Field	Detail
Priority	High
Description	The system shall allow shelter staff to record background/reference check results and move an adoption application through its states (Pending Review → Approved or Rejected; Approved → Completed or Rejected), notifying the applicant of each change.
Use case(s)	UC-03, UC-04, UC-09, UC-07, UC-08
Acceptance criteria	Pass: 'Approved' is accepted only when a background-check result is recorded and no other application for the same animal is Approved, and sets the animal to 'Adoption Pending'; 'Rejected' requires a reason code and returns an 'Adoption Pending' animal to 'Ready for Adoption'; 'Completed' sets the animal to 'Adopted' and closes its other open applications as Rejected; the applicant is notified within 1 minute of each change. Fail: Any change happens without its condition, two applications for one animal are Approved, or a notification is not sent within 1 minute.
Rationale	Formal decision checkpoint; guarantees one adopter per animal and a defined end state ('Adopted').

FR-005 — Foster placement timelines

Field	Detail
Priority	Medium
Description	The system shall allow shelter staff to manage foster home placements: record start date, expected end date and assigned foster caregiver, flag placements past their expected end date, and end a placement.
Use case(s)	UC-06
Acceptance criteria	Pass: Marking an animal 'In Foster' requires a placement with start date, caregiver and an expected end date later than the start date; a placement past its expected end date appears in the staff overdue list; ending a placement restores the animal's previous status ('In Intake' or 'Ready for Adoption'). Fail: A placement is saved with an end date on or before its start date, an 'In Foster' animal has no active placement, an overdue placement is not flagged, or an ended placement leaves the animal 'In Foster'.
Rationale	Problem statement requires managing foster placement timelines; Medium because adoptions can run without fostering.

Priority basis: High = needed for a safe, working adoption cycle (intake, application, decision, data protection); Medium = improves usability or operations, but the adoption cycle still works without it.

5. Non-Functional Requirements

NFR-001 — Catalog performance and exposure

Field	Detail
Type	Non-Functional (Performance & Security)
Priority	Medium
Description	The pet search catalog must support multi-attribute filtering (Species, Age, Breed, Good with Kids / Other Pets) with page response < 200 ms at the 95th percentile, and must expose only public animal fields.
Use case(s)	UC-01
Acceptance criteria	Pass: A load test with 200 concurrent users running mixed filter queries records a 95th-percentile response < 200 ms, and no catalog response contains medical-intake or applicant data. Fail: 95th-percentile response ≥ 200 ms, or any catalog response includes staff-only fields.
Rationale	Given in problem statement (200-user peak load is an assumed figure); Medium because a slower catalog degrades but does not block adoption.

NFR-002 — Access control and privacy

Field	Detail
Type	Non-Functional (Security & Privacy)
Priority	High
Description	The system shall require login for all functions except the public catalog, restrict medical-intake records and background-check results to Shelter Staff, and restrict each adopter questionnaire to Shelter Staff and the applicant who submitted it.
Use case(s)	UC-10, UC-02, UC-04, UC-05
Acceptance criteria	Pass: An unauthenticated request to any protected record returns HTTP 401; an applicant requesting medical, background-check or another applicant's questionnaire data receives HTTP 403; an applicant can view their own questionnaire. Fail: Any user reads or edits data outside these rules.
Rationale	Protects sensitive animal medical data and applicants' personal information.

6. Data and State Model

6.1 Main Entities

Entity	Key attributes
Animal	id, name, species, breed, age, good with kids / other pets, status

Medical Intake Record	animal, vaccinations, medical examination, quarantine flag (open / cleared), logged by, date
Adoption Application	id, animal, applicant, questionnaire answers, status, reason code, timestamps
Background Check Result	application, result, checked by, date
Foster Placement	animal, caregiver, start date, expected end date, actual end date
User	id, role (Applicant or Staff), credentials
Notification	recipient, application, channel (email, in-app), message, sent at

6.2 Status Lifecycles

- Animal: In Intake → Ready for Adoption (complete intake, no open quarantine flag; FR-001) → Adoption Pending (an application Approved; FR-004) → Adopted (application Completed; FR-004). Adoption Pending returns to Ready for Adoption if the approved application is Rejected. In Intake or Ready for Adoption → In Foster (FR-005); ending the placement restores the previous status, so animals can recover in foster care before being cleared.
- Application: Pending Review → Approved or Rejected; Approved → Completed or Rejected (e.g., adopter withdraws). When an application is Completed, all other open applications for that animal become Rejected. Unsent drafts exist only in the applicant's browser (E2).

7. External Interfaces

- User interface: responsive web pages for the catalog, questionnaire and "My Applications", and staff screens for intake, the review queue, decisions and foster placements (including the overdue list). Validation errors are shown inline with entered data preserved (E1).
- Notification interface: email and in-app alerts for each application status change (FR-004).
- Browser storage: local draft of an unsent questionnaire (E2).

8. Traceability

Requirement	Use case(s)	Planned test	Logged defects
FR-001	UC-05, UC-11	TC-FR001	—
FR-002	UC-01	TC-FR002	BUGBPS57-1
FR-003	UC-02	TC-FR003	BUGBPS57-2
FR-004	UC-03, UC-04, UC-09, UC-07, UC-08	TC-FR004	BUGBPS57-3
FR-005	UC-06	TC-FR005	BUGBPS57-4
NFR-001	UC-01	TC-NFR001	—
NFR-002	UC-10, UC-02, UC-04, UC-05	TC-NFR002	BUGBPS57-5

Defects logged on the Jira bug board (6-Software-Testing):

- BUGBPS57-1 — Catalog filter returns no results when 'Good with Kids' is combined with Species
- BUGBPS57-2 — Questionnaire submission fails silently when the animal listing is deleted mid-submission
- BUGBPS57-3 — Staff can approve an application without completing the background check
- BUGBPS57-4 — Foster placement end date allows values earlier than the start date
- BUGBPS57-5 — Unauthenticated users can view medical-intake records via direct URL

Full matrix, including use case → requirement: 1-RE/RTM.xlsx.